

Frequency-Domain Analysis of Thermoacoustic Coupling in a Liquid-Propellant Rocket Engine

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Abstract

Combustion instability, the in-phase coupling of unsteady heat release with a chamber acoustic mode, is among the most destructive failure modes of a liquid rocket engine, and is most dangerous during the ignition transient. This report presents a high-fidelity frequency-domain acoustic model of an LRE chamber and nozzle in which the complex pressure amplitude satisfies the Helmholtz equation on an axisymmetric domain, solved across a 500 to 3000 Hz sweep that brackets the ignition-band modes. Reading the total sound-pressure-level field locates the pressure antinodes relative to the heat-release zone, and the mitigation strategy chooses chamber geometry and damping so that the Rayleigh integral stays negative. The method is a fast screening tool that underpins ongoing work prepared for the IEEE Aerospace Conference.

Keywords: thermoacoustics; combustion instability; liquid rocket engine; Helmholtz equation; frequency domain; Rayleigh criterion

I Introduction

Combustion instability is among the most destructive failure modes of a liquid-propellant rocket engine (LRE). When the unsteady heat release of the flame couples in phase with an acoustic mode of the chamber, the resulting pressure oscillations can grow until they damage the injector face, the chamber wall or the turbomachinery, and the risk is sharpest during the ignition transient, when the acoustic field is establishing itself and the mean state is changing quickly.

This report describes a high-fidelity frequency-domain acoustic model of an LRE chamber and nozzle, built to identify the chamber geometry and damping that suppress these oscillations before they can lock in. Working in the frequency domain makes the analysis fast enough to compare candidate chamber geometries, which is what a conceptual-design screen needs.

II Governing equations

The relevant physics is the response of the chamber acoustics to a harmonic forcing at angular frequency $\omega = 2\pi f$. In the pressure-acoustics, frequency-domain formulation the complex pressure amplitude p satisfies the Helmholtz equation

$$\nabla \cdot \left(\frac{1}{\rho} \nabla p \right) + \frac{\omega^2}{\rho c^2} p = 0, \quad (1)$$

solved here on an axisymmetric representation of the chamber and nozzle (Fig. 1), with the mean density ρ and sound speed c taken from the combustion state. The driver of instability is the Rayleigh criterion: a mode grows when the

heat-release fluctuation q' is, on average, in phase with the pressure fluctuation p' over a cycle,

$$\oint p' q' dt > 0. \quad (2)$$

The engineering objective is therefore to shape the acoustic field, and the damping, so that this integral stays negative at the chamber's natural frequencies.

III Method

The chamber and nozzle geometry is meshed with a resolved near-wall layer (Fig. 2) so that the acoustic response and its wall damping are captured accurately. Equation (1) is then solved over a frequency sweep,

$$f \in \{500, 750, 1000, \dots, 3000\} \text{ Hz}, \quad (3)$$

which spans the chamber's low-order longitudinal and transverse modes and brackets the band in which ignition-phase coupling is most dangerous. The total sound pressure level is mapped through the chamber at each frequency to locate the pressure antinodes and to expose where acoustic energy concentrates (Figs. 3, 4, 5).

IV Results and discussion

The frequency response identifies the chamber modes most susceptible to thermoacoustic coupling and, through the sound-pressure-level field, where their antinodes sit relative to the injector and the combustion zone. As the forcing frequency is swept, the antinode pattern migrates through the chamber, so a mode that is benign at one frequency can place a strong antinode over the heat-release region at another. That information drives the mitigation strategy: choosing chamber proportions and damping so that the susceptible antinodes do not overlap the region of strongest heat release, which spatially decouples p' from q' and pushes the Rayleigh integral negative during ignition. Because it is a frequency-domain screen, the method is fast enough to compare chamber geometries before committing to a fully coupled reacting-flow simulation.

V Conclusion

A frequency-domain Helmholtz model of the chamber acoustics, swept across the ignition-band modes and read through the sound-pressure-level field, gives a practical route to choosing chamber geometry and damping that suppress thermoacoustic instability. The method underpins ongoing work prepared for the IEEE Aerospace Conference.

References

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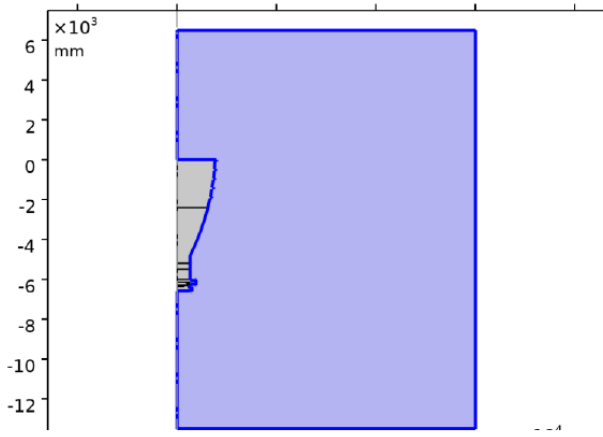


Figure 1. Axisymmetric acoustic domain of the chamber and nozzle.

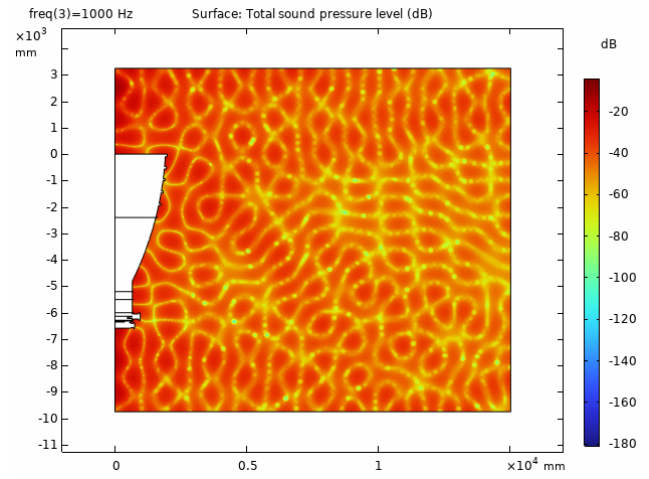


Figure 3. Total sound pressure level at the low end of the ignition band.

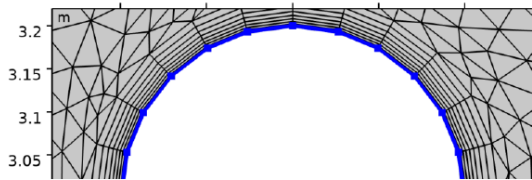


Figure 2. Resolved near-wall mesh used to capture the acoustic response and wall damping.

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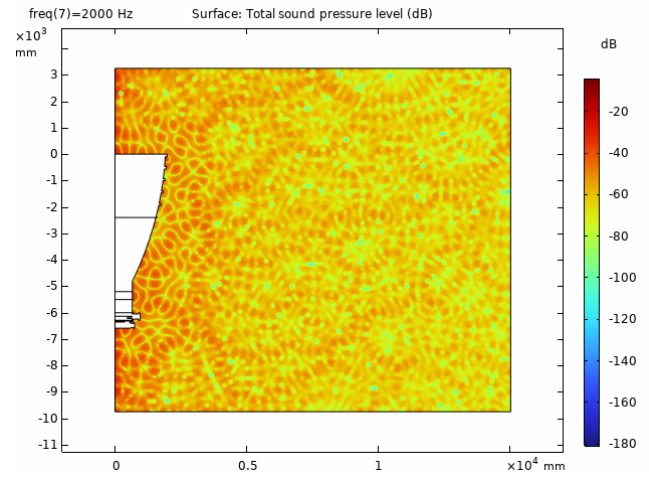


Figure 4. Total sound pressure level at a mid-band frequency.

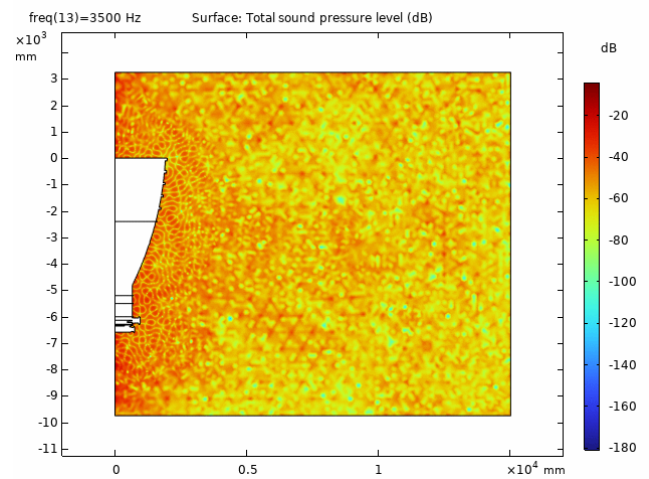


Figure 5. Total sound pressure level at the high end of the ignition band; the antinode pattern has migrated.